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For correspondence:

zhchi@ustc.edu.cn

rjzhang@ustc.edu.cn

jhyuan@ustc.edu.cn

* These authors contributed equally.

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Enhanced bacterial chemotaxis in confined microchannels: Optimal performance in lane widths matching circular swimming radius

Caijuan Yue*, Chi Zhang*, Rongjing Zhang, Junhua Yuan

Hefei National Research Center for Physical Sciences at the Microscale, and Department of Physics, University of Science and Technology of China, Hefei, China

eLife Assessment

This **important** work examines the effects of side-wall confinement on chemotaxis of swimming bacteria in a shallow microfluidic channel. The authors present **convincing** experimental evidence, combined with geometric analysis and numerical simulations of simplified models, showing that chemotaxis is enhanced when the distance between the side walls is comparable to the intrinsic radius of chiral circular swimming near open surfaces. This study should be of interest to scientists specializing in bacteria-surface interactions.

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Abstract

Understanding bacterial behavior in confined environments is helpful for elucidating microbial ecology and developing strategies to manage bacterial infections. While extensive research has focused on bacterial motility on surfaces and in porous media, chemotaxis in confined spaces remains poorly understood. Here, we investigate the chemotaxis of *Escherichia coli* within microfluidic lanes under a linear concentration gradient of L-aspartate. We demonstrate that *E. coli* exhibits significantly enhanced chemotaxis in lanes with sidewalls compared to open surfaces. We attribute this phenomenon primarily to the intrinsic clockwise circular motion of surface-swimming bacteria and the subsequent alignment effect upon collision with the sidewalls. By varying lane widths, we identify that an 8 μm width—approximating the radius of bacterial circular swimming on surfaces—maximizes chemotactic drift velocity. These results are supported by both experimental observations and stochastic simulations, establishing a clear proportional relationship between optimal lane width and the radius of bacterial circular swimming. Further geometric analysis provides an intuitive understanding of this phenomenon. Our results may offer insights into bacterial navigation in complex biological environments such as host tissues and biofilms, providing a preliminary step toward exploring microbial ecology in confined habitats and potential strategies for controlling bacterial infections.

Introduction

Microorganisms inhabit a variety of complex confined environments. Sperm swimming in the reproductive tract of a female animal (1), bacteria navigating through host cells and tissues (2), cells moving in the wrinkle of biofilm (3), and swarm cells maneuvering in crowded cell clusters (4) are all examples of organisms operating in geometrically confined environments. Furthermore, many diseases have been found to be related to bacterial motion in confinement, such as urinary tract infections and neonatal sepsis.

Previous research has explored the motion of bacteria in confined environments. Studies have shown that flagellated bacteria tend to accumulate near both solid-liquid (5, 6) and air-liquid interfaces (7), and exhibit a specific direction of circular swimming (8). For example, *E. coli* swim clockwise when observed from above a solid surface (8–11), whereas *Caulobacter crescentus* move in tight, counter-clockwise circles when viewed from the liquid side. Moreover, Tang *et al.* found that flagellated bacteria orbit within the thin fluid film around micrometer-sized particles (12). Bacteria also display different motion behaviors in porous media and tubes (13–19), as well as in other types of confinement (20–22). In narrow channels with width less than 4 μm , bacterial growth (23), swimming trajectories (24) and swimming speeds (25) are significantly impacted compared to bulk conditions. Furthermore, studies on bacterial transport in rectangular microchannels under flow have demonstrated that confinement influences both bacterial fluxes and spatial distribution, with cells exhibiting distinct transport dynamics along channel edges (26). However, while motility in confinement is well-documented, the specific rules governing bacterial chemotaxis within geometrically constrained environments remain less understood.

E. coli can sense and move in chemical gradients through a set of chemotactic proteins (27, 28). The transmembrane chemoreceptors detect attractants or repellents and transmit signals into the cell by modulating the autophosphorylation of the histidine kinase CheA. Attractant binding suppresses CheA autophosphorylation, while repellent binding promotes it. This modulation alters the concentration of the phosphorylated response regulator protein, CheY-P (28–31). CheY-P binds to the cytoplasmic domain of flagellar motors, altering the probability of their clockwise rotation (32, 33). Clockwise and counterclockwise rotations of motors lead to reorientation (tumble) and smooth swimming (run) modes of the cell body, respectively (34). The adaptation enzymes CheR and CheB methylate and demethylate the receptors, respectively, mediating sensory adaptation (35, 36).

While the molecular mechanism is well-characterized, the physical environment plays a crucial role in chemotactic efficiency. Bacterial chemotaxis has been observed under various degrees of confinement, often yielding complex behaviors. For instance, in quasi-2D microchannels with a depth of 2 μm , bacteria were found to adjust their run lengths and tumble frequencies to respond to chemical gradients (37). Similarly, studies in 10 μm -depth microchambers revealed that *E. coli* maintains chemotactic performance in the presence of evenly spaced obstacles (38). While these studies focused on confinement between top and bottom surfaces, recent work by Braham *et al.* showed that bacteria on a single solid surface can exhibit an escape response to high-concentration chemorepellents (250 mM $\text{Ni}(\text{NO}_3)_2$) (39). Conversely, other research has shown that bacterial chemotactic drift velocity decreases sharply near sample chamber surfaces compared to bulk liquid, a phenomenon attributed to circular cell movement at these interfaces (40). This observation suggests that a single surface can significantly impair bacterial chemotactic performance. Yet, a study of chemotaxis in two-dimensional bacterial swarms on semisolid agar surfaces has demonstrated that *E. coli* can exhibit effective chemotaxis in attractant gradients, especially at high cell densities (4). This apparent contradiction is likely due to cell-cell interactions mitigating surface effects in dense populations. Despite these investigations into motility under surface effects, a comprehensive understanding of how specific confinement geometries modulate chemotaxis remains limited.

Here, we study the chemotaxis of *E. coli* on surfaces within lanes of different widths, exposed to a linear concentration gradient of L-aspartate using a hydrogel-based microfluidic device. Our findings reveal enhanced chemotactic performance in the presence of sidewalls compared to their absence. Cells in the middle area (MA) exhibited no drift and did not contribute to overall chemotaxis. However, cells near the left sidewall (LSW) and right sidewall (RSW) demonstrated negative and positive drift, respectively, with the positive drift velocity from the RSW exceeding the negative drift velocity from the LSW.

Furthermore, by comparing lanes of different widths within the same stable linear gradient field, we observed that cells exhibited the highest drift velocity in 8 μm -wide lanes, which approximates the peak value in the distribution of circular swimming radii on surfaces. This optimal width for maximal chemotaxis resulted from the largest percentage of cells occupying the right sidewalls.

Stochastic simulations of bacterial chemotaxis corroborated that the optimized lane width varies with the bacteria's circular swimming radius. Our findings have important implications for the screening of swimming organisms and the study of chemotactic behavior in real-world scenarios.

Results

Chemotaxis of *E. coli* within lane confinements

To study bacterial chemotaxis in confined environments such as tubes or interstitial tissues, we developed a hydrogel-based microfluidic device featuring lanes of different widths. The device design, illustrated in Fig. 1A [↗](#), incorporates two 2% (w/v) agarose walls inserted between three main channels: a sink channel, a cell channel, and a source channel. These walls prevent cell passage while allowing chemical diffusion. The agarose walls and main channels measure 100 μm and 400 μm in width, respectively, with the device having a depth of 30 μm .

We established a linear concentration gradient (0.05 $\mu\text{M}/\mu\text{m}$) of L-aspartate, an attractant sensed by the Tar receptor, by flowing motility medium through the sink channel and 50 μM L-aspartate solution (prepared in motility medium) through the source channel at a constant rate of 10 $\mu\text{L}/\text{min}$. The resulting gradient across the cell channel is depicted in Fig. S1.

The lanes of various widths were created using parallel PDMS (polydimethylsiloxane) pillars spaced at different distances. Each lane, approximately 160 μm in length, is highlighted by a yellow rectangular box in Fig. 1A [↗](#). *E. coli* cells, confined to the cell channel, sensed the attractant gradient and performed chemotaxis within these lanes of different widths.

To enhance imaging contrast, we introduced a plasmid expressing mCherry fluorescent protein (pTrc99a-mcherry) into wild-type *Escherichia coli* HCB1, denoting it as HCB1-pTrc99a-mcherry. A high-sensitivity CMOS camera was used to record bacterial movement trajectories on the bottom surfaces of the lanes in the cell channel, where most cells accumulate due to the combined effects of gravity and hydrodynamic wall attraction in the Percoll-free medium.

We recorded the movement of HCB1-pTrc99a-mcherry cells on the liquid-glass surface of a 44- μm -wide lane with and without a L-aspartate concentration gradient, respectively. An example video is shown in Movie S1, and several representative trajectories are illustrated in Fig. 1B [↗](#).

Considering the potential escape into the third dimension at the edge [\(26\)](#), we calculated the drift velocities v_d of the trajectories from both assays along the x -direction (up-gradient) by fitting the relation $\langle x(i+n) - x(i) \rangle \sim n\Delta t$ ($n=1, 3, 5, \dots, 39$) with a linear function, the slope of which is the drift velocity. Here, $x(i)$ represents the cell's x -position in the i -th frame, and Δt denotes the time interval between two frames (0.05 s). This method yields a drift velocity that is independent of potential cell loss into the third dimension. As shown in Fig. 1C [↗](#), the drift velocity v_d was 1.6 ± 0.3 $\mu\text{m}/\text{s}$ and 0.2 ± 0.2 $\mu\text{m}/\text{s}$ (mean $\pm$ SD) for the gradient and control (no-gradient) assays, respectively. Thus, bacteria in the L-aspartate gradient exhibited obvious chemotaxis on the lane surfaces, contrasting with previous findings on liquid-solid surfaces without sidewalls [\(40\)](#).

The cells in the right sidewall region dominated the chemotaxis of *E. coli* within lane confinements

To elucidate the mechanism of bacterial chemotaxis within lane confinements, we analyzed cell trajectories, identifying three distinct motion states due to the presence of sidewalls. As illustrated in Fig. 2A [↗](#), cells exhibited straight-line swimming along sidewalls, movement towards or away from sidewalls, or circular swimming. Fig. 2B [↗](#) displays three representative trajectories. Circular swimming is characteristic of typical *E. coli* surface motion [\(8\)](#). We further analyzed the trajectories of bacteria colliding with the sidewalls. As shown in Fig. S2, cells colliding with the sidewalls tend to align with and swim along the sidewalls, and the alignment timescale is negligible compared to the typical wall residence time [\(6\)](#).

We distinguished these motion states by calculating the rotational exponent (γ_R) for each trajectory, fitting the mean-squared orientational displacement (MSOD) for $\tau < 0.4$ s:

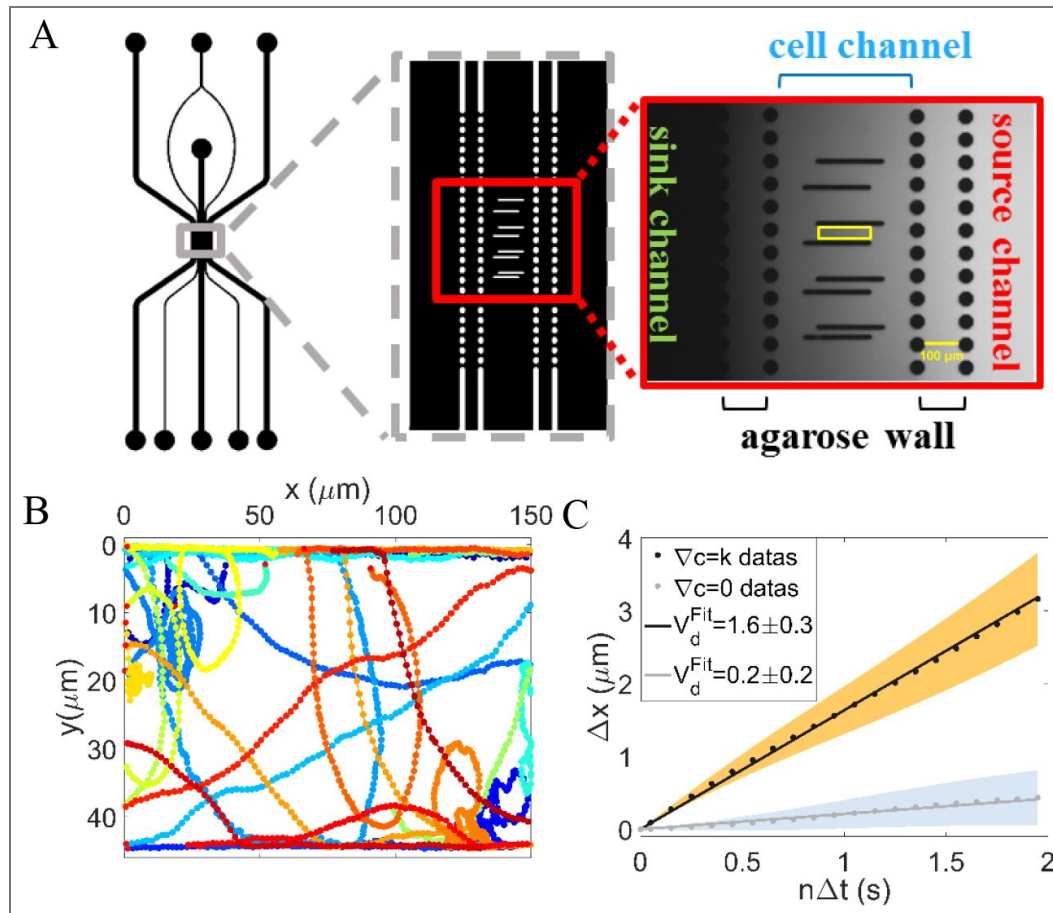


Fig. 1.

A. Microfluidic device used in the experiment (top view). **B.** Thirty bacterial trajectories selected from the data of 44-μm-wide lane in gradient assays. These represent a subset of the trajectories analyzed in panel **C**. **C.** The relationship between Δx and $n\Delta t$ calculated from all trajectories in the lanes with a width of 44 μm. The black dots represent the gradient assay ($\nabla c = 0.05$ μM/μm) with a total of 3206 tracks from 6 movies, while the gray dots represent no gradient ($\nabla c = 0$) with a total of 4755 tracks from 10 movies. The light-yellow and light-blue shadows represent the standard error of the mean (SEM) of different trajectories in the gradient assays and the control assays, respectively. Linear fitting was performed to obtain $V_d = 1.6 \pm 0.3$ μm/s (black solid line) and $V_d = 0.2 \pm 0.2$ μm/s (gray solid line) for gradient and control assays respectively. Error in drift velocity represents standard deviation.

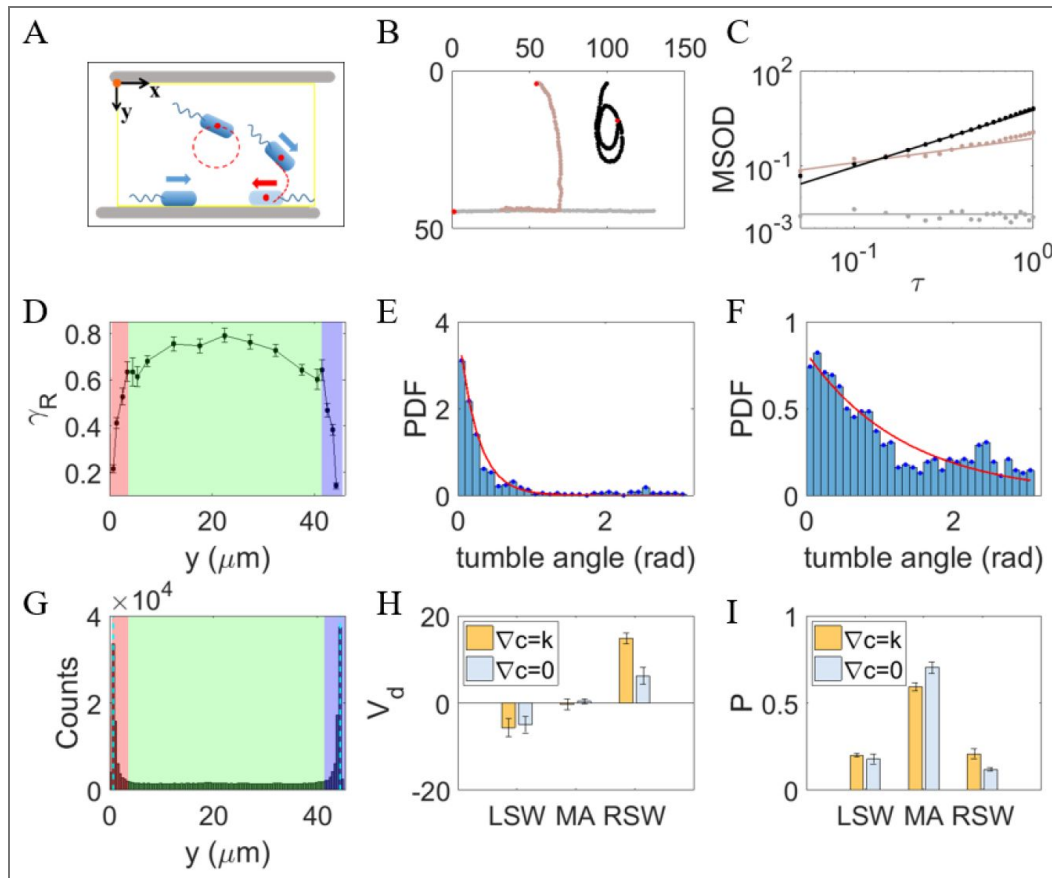


Fig. 2.

A. Schematic drawing of three typical motion states. **B.** Trajectories for three typical motion states. Red dots represent the start of each trajectory. **C.** Calculating the rotational exponent for each trajectory, γ_R , by fitting the mean-squared orientational displacement (MSOD). Dots were experimental data calculated from the tracks in Fig. 2B. Solid lines were fitting results with $\langle \text{MSOD} \rangle = C_R \tau^{\gamma_R}$. The fitted γ_R values are 0, 0.77, and 1.80 for gray, brown, and black tracks, respectively. **D.** Relationship between the rotational exponent γ_R and the mean y position of the bacterial trajectories. Errors denote SEM. Red and purple shaded area represent the region 3 μm from the LSW and RSW, respectively. Green shaded area represents the MA region. **E&F.** Normalized distribution of tumble angle for tracks along the sidewalls (**E**) and in the MA region (**F**). Red solid lines are fitting results with an exponential function: $y = a * \exp(x/b)$. **G.** Distribution of bacteria along the y -axis in the lanes with a width of 44 μm . The shades of different colors denote the same meaning as in Fig. 2D. **H.** Drift velocity of bacterial cells in the three regions. **I.** Proportions of bacterial trajectories in the three regions. Errors are standard deviations (SD).

$$\langle \text{MSOD} \rangle = \langle (\theta(t + \tau) - \theta(t))^2 \rangle = C_R \tau^{\gamma_R},$$

where $\theta(t)$ is the angle between the swimming direction and the positive x -axis direction at time t , τ is the lag time, C_R is a constant, and γ_R is the rotational exponent. Fig. 2C shows the γ_R values for the three trajectories in Fig. 2B, while Fig. 2D illustrates the relationship between γ_R and average y -coordinate. Trajectories further from sidewalls exhibit larger γ_R , indicating more curved paths, while γ_R decreases sharply as the cells approach sidewalls, representing straighter trajectories.

We also analyzed tumble kinematics for cells swimming along and away from sidewalls. Fig. 2E and 2F show normalized tumble angle distributions for cells in sidewalls (SW) and middle areas (MA), respectively. Red solid lines represent exponential function fits $y = a * \exp(x/b)$, with fitted b values of 0.25 rad (SW) and 1.35 rad (MA). Consistent with previous reports (41), we observed a higher occurrence of large reorientation angles for surface-swimming cells in the MA region (Fig. 2F). Interestingly, we also found large tumble angles in the SW region, albeit with a smaller characteristic angle than in the MA region. Movie S2 demonstrates an example of a large reorientation angle during tumbling.

Given the strong spatial dependence of trajectory characteristics, we classified the cells into three distinct populations. The distribution of cells along the y -axis is shown in Fig. 2G. Based on mean distances from the sidewalls, we divided the lane into three areas (viewing along the positive x -axis): the left sidewall (LSW), the middle area (MA), and the right sidewall (RSW). As shown by different shaded colors in Fig. 2D and 2G, we defined cells in LSW, RSW and MA for trajectories with $\langle x \rangle \leq d$, $\langle x \rangle \geq w - d$, and $d < \langle x \rangle < w - d$, respectively. The threshold d was set at 3 μm (approximately the average length of a cell body), and w represents the lane width. The proportion of trajectories in different regions is shown in Fig. 2I. For the gradient assay, the probabilities of trajectories being in LSW, MA and RSW regions were 0.20 ± 0.01 , 0.59 ± 0.02 , and 0.21 ± 0.03 , respectively. In the control assay without gradient, these values were 0.18 ± 0.03 , 0.70 ± 0.03 , and 0.12 ± 0.01 , respectively.

We then compared the drift velocity of cells in the three areas separately for both gradient and control assays. As shown in Fig. 2H, cells in the LSW region tended to display a negative drift with $v_d^{\text{LSW}} = -5.7 \pm 2.1 \mu\text{m/s}$ and $-5.0 \pm 2.0 \mu\text{m/s}$ for gradient and control assays, respectively. Cells in the MA region exhibited minimal drift with $v_d^{\text{MA}} = -0.3 \pm 1.2 \mu\text{m/s}$ and $0.4 \pm 0.6 \mu\text{m/s}$ for gradient and control assays, respectively. Cells in the RSW region displayed a typical positive drift with $v_d^{\text{RSW}} = 14.8 \pm 1.2 \mu\text{m/s}$ and $6.2 \pm 1.9 \mu\text{m/s}$ for gradient and control assays, respectively. Thus, cells in the RSW region dominated the difference in drift velocity between the two assays.

In summary, cells swim on the right-hand side under the effect of the bottom surface of lanes (9). The LSW and RSW cells swim down and up the gradient, respectively, while MA cells swim in circles without any drift. RSW cells demonstrated more persistent swimming towards the attractant in the L-aspartate gradient, exhibiting a higher positive drift velocity. Detailed information on the difference in drift velocity between LSW and RSW cells can be found in the supplemental materials. Notably, the attractant gradient minimally affects the ratio of bacteria in different regions.

The dependence of chemotactic performance on lane width

Having established that *E. coli* chemotaxis on lane surfaces is closely related to sidewall confinement, we investigated chemotaxis in lanes of different widths to mimic bacterial chemotactic behavior in interstitial tissues or tubes of varying sizes. We compared five lanes with widths of 6 μm , 8 μm , 10 μm , 25 μm , and 44 μm . Drift velocity was calculated using the same method as in Fig. 1C, with results shown in Fig. 3A. The mean drift velocities were $1.8 \pm 1.3 \mu\text{m/s}$, $7.5 \pm 1.4 \mu\text{m/s}$, $4.6 \pm 1.7 \mu\text{m/s}$, $2.5 \pm 0.5 \mu\text{m/s}$, and $2.6 \pm 0.4 \mu\text{m/s}$ for lanes of width 6 μm , 8 μm , 10 μm , 25 μm , and 44 μm , respectively. Bacteria exhibited optimal chemotaxis (highest drift

velocity) in lanes that were 8 μm wide. To rule out any potential ratchet effect caused by the asymmetric entrances, we performed control measurements using a setup with straight entrances; these results confirmed our findings and are shown in Fig. S3.

Noting that this optimal width is of the same order of magnitude as the radius of bacterial circular swimming on a solid surface, we measured the radius of circular swimming for cells in the MA region. As shown in Fig. 3B, the peak radius in the distribution of circular swimming radii is $\sim 10 \mu\text{m}$, which is close to the lane width supporting optimal chemotaxis.

It has been reported that bacteria swimming in capillaries of different radii possess different swimming speeds (42). Therefore, we calculated the cell swimming speeds for all lane widths. As shown in Fig. 3C, swimming speed remains relatively constant across different lane widths. This indicates that the optimized chemotaxis is not a result of varying swimming speeds.

The optimal chemotaxis resulted from more cells moving towards the attractant along the right sidewall

To identify the underlying mechanism of optimal lane width for chemotaxis, we evaluated the drift velocity for cells in the LSW, MA, and RSW regions separately. The results are shown in Fig. 4A. Cells in the MA region did not exhibit effective drift, while LSW and RSW cells demonstrated negative and positive drift, respectively, consistent with the findings in Fig. 2H. The drift velocity for RSW cells increased with lane width, and remained relatively constant above a lane width of 8 μm , thus not exhibiting a highest value in the 8 μm -wide lanes. Therefore, the width-dependent optimal chemotaxis is not attributable to an optimal drift velocity for RSW cells.

We then sought a comparison of cell proportions in the three areas, as plotted in Fig. 4B. The proportion of cells in the MA region increased gradually with lane width due to the increase in area. Notably, the largest proportion of RSW cells appears in the 8 μm -wide lane. In other words, the lane width-dependent optimal chemotaxis primarily resulted from changes in cell numbers on the sidewalls. The relation between cell enrichment and lanes' width was plotted in Fig. S4.

To further investigate changes in up- and down-gradient cell numbers in the RSW region, we calculated the proportion of RSW-UG (cells swimming up-gradient in the RSW) and RSW-DG (cells swimming down-gradient in the RSW) cells. As shown in Fig. 4C, more cells moved towards the attractant in the 8 μm -wide lane. Thus, the optimal chemotaxis in the 8 μm -wide lane resulted from a higher number of cells moving towards the attractant along the right sidewall.

Simulation of *E. coli* chemotaxis within lane confinements

To explore the relationship between the circular swimming of bacteria on lane surfaces and optimal chemotaxis, we simulated two-dimensional chemotaxis of bacteria with varying circular swimming radii on lanes of different widths in a steady linear gradient of L-aspartate.

In our simulation, cells were treated as self-propelled particles performing a random walk in a run-and-tumble mode within a two-dimensional space. Cells could be in either a run or tumble state, determined by the intracellular chemotaxis signal. During a run, cells swam smoothly at a constant speed of $v_0 = 20 \mu\text{m/s}$. The effect of the bottom surface on bacterial swimming was modeled by adding a constant angular velocity term to the swimming direction:

$$\theta(t + \Delta t) = \theta(t) + \left(-\frac{v_0}{r}\right) \Delta t + \sqrt{2D_r \Delta t} n(0, 1)$$

where θ denotes the angle between the swimming direction and the positive x-axis, Δt is the time step, r is the radius of circular swim, D_r is the rotational diffusion coefficient, and $n(0, 1)$ represents a random number drawn from a standard normal distribution. When cells collided with sidewalls, their velocities instantly aligned with the sidewall (43), and could potentially leave the sidewall after their next tumble. During a tumble, cells stopped moving, and their swimming direction changed. The change in θ was selected from the distribution of tumble angles observed in Fig. 2E and F for different areas.

Fig. 3.

A. The drift velocity for the experiments in lanes of different widths. The error in drift velocity is the standard deviation. **B.** The distribution of the radius of circular swimming from 382 trajectories in the MA regions. The peak value of the radius is ~ 10 μm . **C.** The swimming speed of cells in lanes with different widths. The mean swimming speeds and standard deviations for lanes with widths of 6 μm , 8 μm , 10 μm , 15 μm , 25 μm , and 44 μm are 22.7 ± 8.7 $\mu\text{m/s}$, 25.4 ± 7.7 $\mu\text{m/s}$, 27.1 ± 7.9 $\mu\text{m/s}$, 21.9 ± 7.8 $\mu\text{m/s}$, 22.1 ± 7.7 $\mu\text{m/s}$, and 22.6 ± 7.5 $\mu\text{m/s}$, respectively.

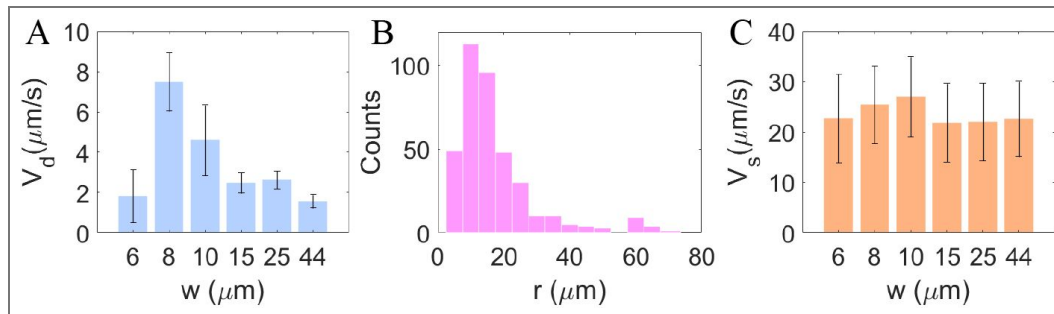
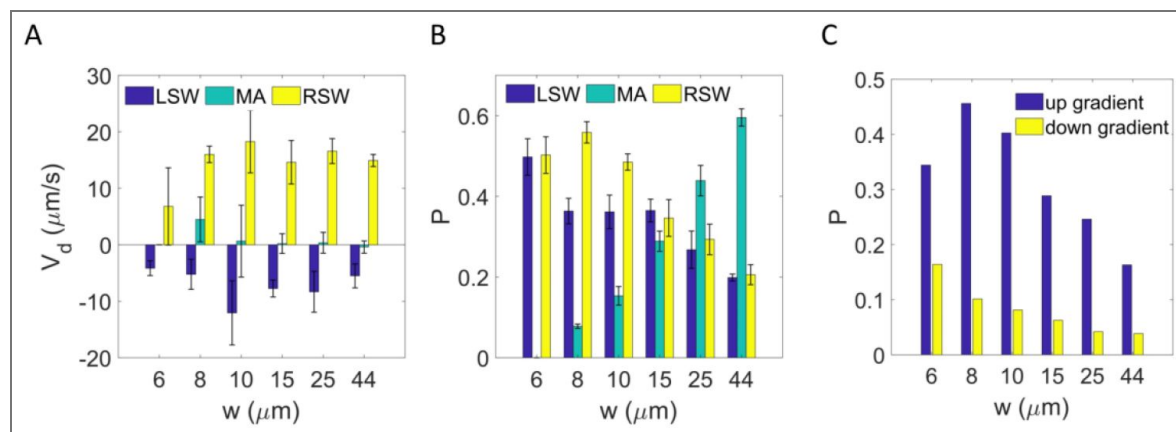


Fig. 4.

A. Mean drift velocities in the LSW, MA, and RSW regions of lanes with different widths. The errors are SD. **B.** The proportion of bacterial cells in the LSW, MA, and RSW regions of lanes with different widths. Errors represent SD. **C.** The proportion of RSW-UG (cells swimming up-gradient in the RSW) and RSW-DG (cells swimming down-gradient in the RSW) for lanes with different widths.



Initially, 100 cells were evenly distributed on the surface of a lane measuring 160 μm in length and w in width. A linear concentration gradient of L-aspartate, identical to the experimental conditions, was established along the x -axis according to:

$$c(x) = 0.05x + 22,$$

with the concentration c in μM and the position x in μm . The simulation for each set of parameters ran for 150 s with a timestep of 0.05 s and was repeated 50 times.

Figure 5A illustrates the relationship between chemotactic drift velocity and lane width for cells with a 10 μm -radius circular swim. The highest drift velocity was observed at a lane width of 8 μm . Figure 5B shows the proportions of cells in the three regions (LSW, MA, RSW) for different lane widths, presenting profiles similar to those observed experimentally in Fig. 4B. The drift velocity for 3 regions was shown in Fig. S5. Results for different swimming radii are presented in Fig. 5C, demonstrating that the optimal lane width for chemotaxis is closely related to the radius of circular swimming. It is important to note that our model treats bacteria as active particles and does not explicitly account for the steric exclusion of the flagella and cell body. In reality, these effects would restrict the cells' ability to turn freely in narrow lanes ($\leq 6 \mu\text{m}$), thereby reducing the cell density mismatch between the LSW and RSW and decreasing the drift velocity compared to the simulation.

As shown in Fig. 5D, the relationship between optimal lane width (w) and circular swimming radius (r) was fitted with a linear function $w = kr$, with $k = 0.66 \pm 0.03$. This indicates a clear proportional relationship between the lane width for optimal chemotaxis and the circular swimming radius of bacteria.

Geometrical analysis of optimal lane width for chemotaxis

We have established that the optimal lane width for chemotaxis (8 μm) primarily results from a higher proportion of cells in the RSW region swimming up-gradient. Additionally, this optimal width is directly proportional to the circular swimming radius of the bacteria. To further elucidate the underlying mechanism of this phenomenon, we employed geometrical analysis.

We assumed an initial uniform distribution of cells across the lane width. As shown in Fig. 6A-C, we divide the width of the lane into three scenarios:

1. $0 < w \leq r$: For a cell at any point P in the channel, its swimming direction can be arbitrary. However, only cells swimming in directions between θ_1 and θ_2 (highlighted in green) can swim up-gradient in the RSW after their first collision with the sidewall. The proportion of these cells can be calculated as:

$$\begin{aligned} \frac{1}{w} \int_0^w \frac{\theta_1 + \theta_2}{2\pi} dy &= \frac{1}{2\pi w} \int_0^w dy \left(\arccos \frac{r-w+y}{r} + \arccos \frac{y}{r} \right) \\ &= \frac{1}{2\pi m} \left[-(1-m) \arccos(1-m) + \sqrt{1-(1-m)^2} + m \arccos m \right. \\ &\quad \left. - \sqrt{1-m^2} + 1 \right], \end{aligned}$$

where $m = w/r \leq 1$.

2. $r < w \leq 2r$: Only the cells in the region $y < r$ and with velocity direction between θ_1 and θ_2 can swim up-gradient in the RSW after their first collision with the sidewall. The proportion of cells in this region is:

$$\frac{1}{w} \int_0^r \frac{\theta_1 + \theta_2}{2\pi} dy = \frac{1}{2\pi w} \int_0^r dy \left(2 \arccos \frac{y}{r} \right) = \frac{1}{\pi m}$$

3. $w > 2r$: The proportion of cells that can swim up-gradient in the RSW is calculated the same as in case 2:

$$\frac{1}{w} \int_0^r \frac{\theta_1 + \theta_2}{2\pi} dy = \frac{1}{\pi m}$$

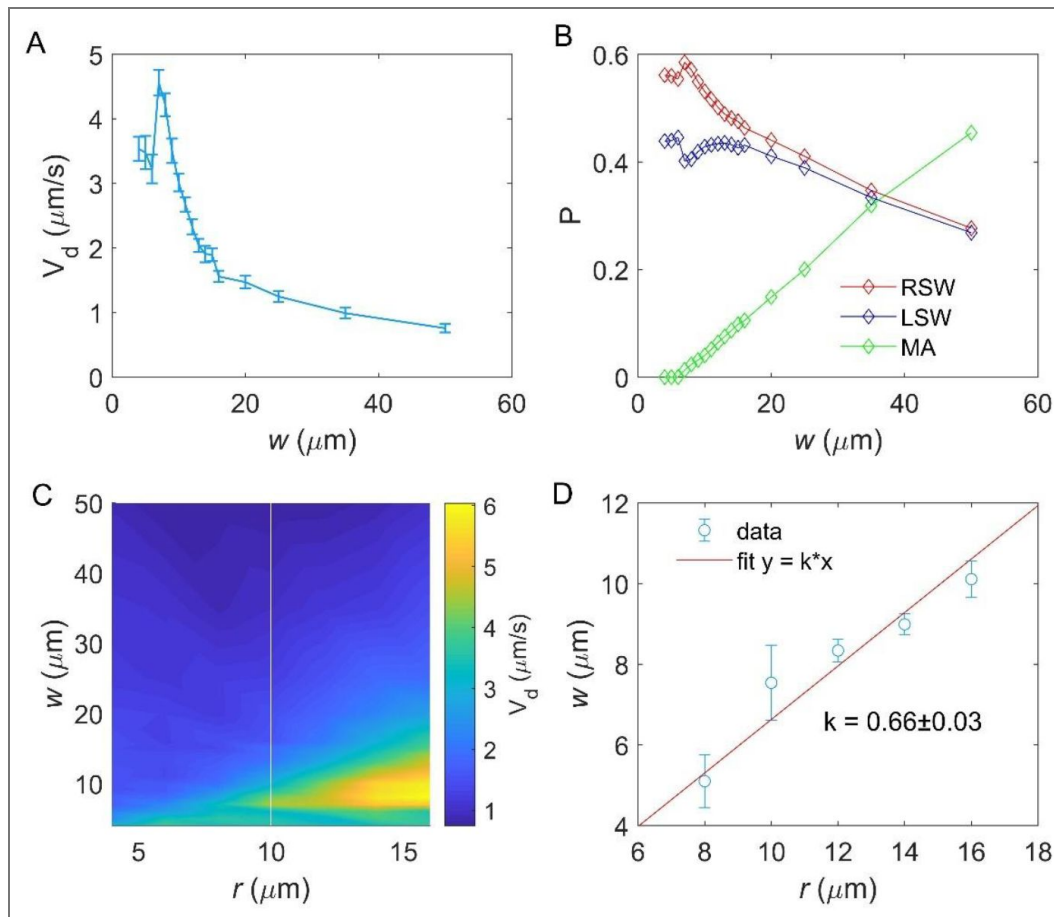


Fig. 5. Simulation of bacterial chemotaxis on the surface of lanes.

A. The relationship between drift velocity V_d of cells with 10 μm -radius circular swim and the lane width w . Errors denote SEM. **B.** The proportion of bacterial trajectories in the LSW, MA, and RSW regions of lanes with different widths, for cells with 10 μm -radius circular swim. **C.** The drift velocity V_d of cells with different radii of circular swim in lanes of different widths. **D.** The relationship between circular swim radius and lane width for optimal chemotaxis (maximal drift velocity) in Fig. 5C. The widths were extracted by fitting the peak value of w in Fig. 5A with a gaussian function. The red solid line represents a linear fit. The slope is 0.66 ± 0.03 . Errors denote SD.

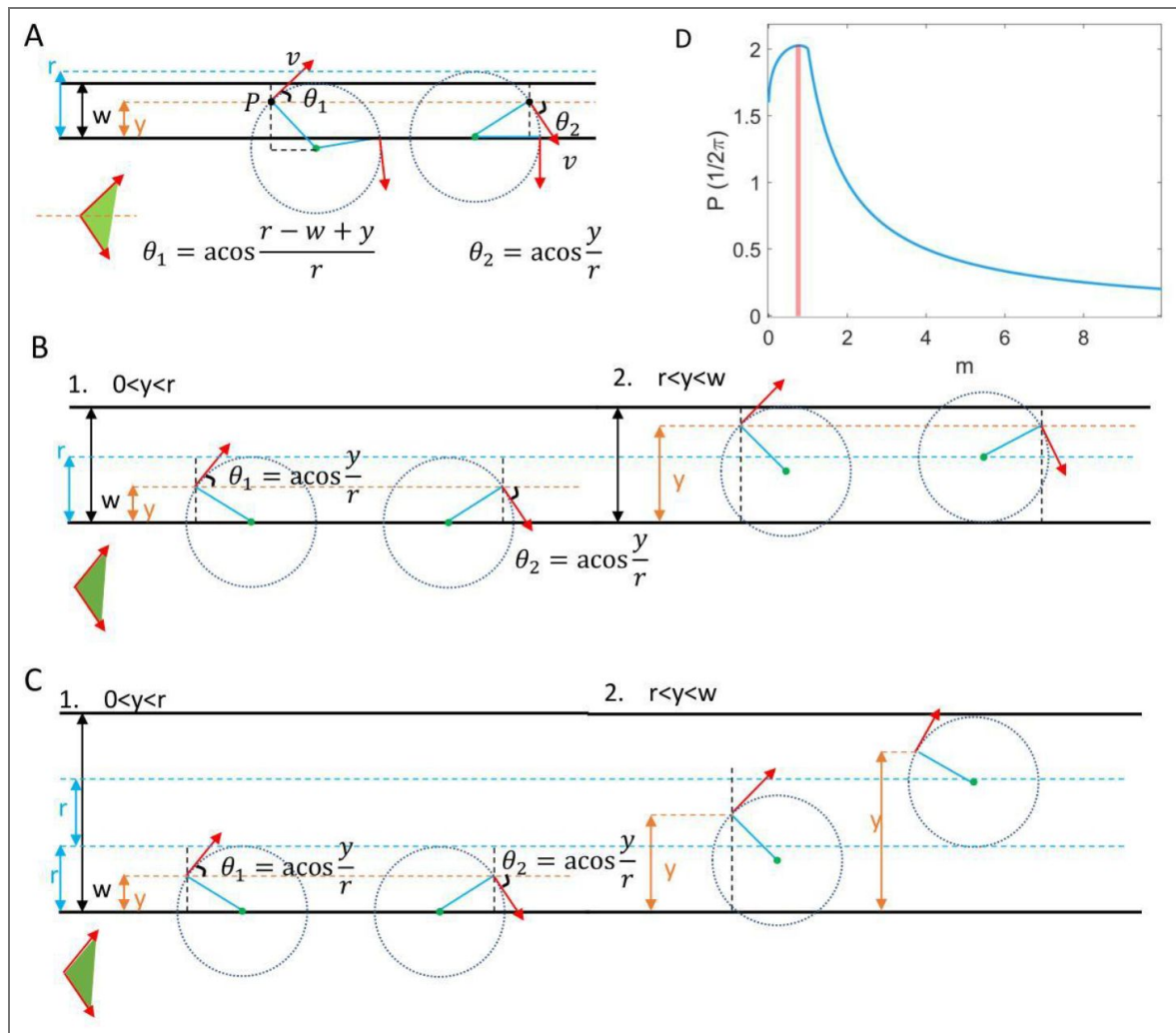


Fig. 6. Geometrical analysis of optimal lane width for chemotaxis.

A. Case 1: $0 < w \leq r$. **B.** Case 2: $r < w \leq 2r$. **C.** Case 3: $w > 2r$. The red arrows denote the velocity direction. The black solid lines represent the sidewalls. Dashed circles are trajectories. The up-gradient direction is along the positive x-axis. The green shade areas label the swim direction of cells that can swim up-gradient in the RSW. **D.** Relationship between probability (P) of cells swimming up-gradient in the RSW and m , where $m = w/r$. Red shaded area denotes m for maximal P , $m \in (0.7, 0.8)$.

The complete relationship between the proportion of cells that can swim up-gradient in the RSW (P) and m is plotted in Fig. 6D [\[10\]](#). The maximum P occurs when $m \in (0.7, 0.8)$, marked by the red shaded area. This value is consistent with the slope k in Fig. 5D [\[10\]](#) and results in the optimal width of about $8 \mu\text{m}$ in Fig. 3A [\[10\]](#) for a circular swimming radius of $10 \mu\text{m}$.

Crucially, the observed asymmetry arises because the angular ranges that geometrically facilitate RSW-UG accumulation (Fig. 6A-C [\[10\]](#)) coincide with the up-gradient direction. Because cells moving up-gradient experience suppressed tumbling, they can maintain the steady circular trajectories required to reach and align with the RSW. Conversely, while a pure geometric analysis suggests a symmetric potential for LSW-DG accumulation, these trajectories coincide with the down-gradient direction. Consequently, these cells experience enhanced tumbling, which distorts the circular trajectories. This prevents them from effectively reaching the LSW and also increases the probability of them leaving the wall.

Discussions

Bacterial behavior in confined environments differs significantly from that in free environments. Previous studies have reported that cells tend to swim in circles and accumulate near a solid-liquid interface, leading to diminished chemotaxis. However, recent research on chemotaxis of swarm cells suggests that crowded environments may enhance bacterial chemotaxis on surfaces. In this study, we examined the chemotaxis of *E. coli* on the bottom surface of microfluidic lanes, which simulate the spatial constraints imposed by neighboring cells in a swarm, exploring the effects of lane confinement on bacterial chemotaxis.

We found that in the presence of sidewalls, cells that would otherwise have been unable to exhibit chemotaxis showed significant chemotactic behavior in a linear gradient of L-aspartate. Cells in different regions of the lanes demonstrated distinct motion behaviors. In the MA region, cells showed no drift, similar to observations in chemotaxis near surfaces. Conversely, cells exhibited positive and negative drift when in the RSW and LSW regions, respectively.

Further investigation into chemotaxis in lanes of varying widths demonstrated that cells exhibited the highest chemotactic drift velocity in lanes that were $8 \mu\text{m}$ wide. This width is close to the radius of bacterial swimming circles. By analyzing the proportion and drift velocity of cells in different regions, we determined that this optimal lane width was primarily due to the higher proportion of cells swimming up-gradient along the right sidewall.

To explore the relationship between chemotactic drift velocity and circular swimming radius of bacteria, we conducted systematic simulations of *E. coli* chemotaxis with different circular swimming radii in lanes of varying widths under a linear concentration gradient of L-aspartate. These simulations explicitly incorporated the clockwise circular motion of surface-swimming bacteria and the alignment effect induced by sidewall collisions. We found that optimal chemotaxis occurred when the swimming radius was close to the lane width, with $w = 0.66r$. This suggests a significant proportional relationship between the optimal lane width for chemotaxis and the circular swimming radius of bacteria.

To elucidate the mechanism underlying this phenomenon, we employed geometric analysis. We demonstrated that bacteria in lanes could swim up-gradient in the RSW with maximal probability when the lane width was 0.7-0.8 times the radius of circular swimming. This finding is consistent with the experimental results presented in Fig. 4C [\[10\]](#).

Our findings have important implications for the screening of single-celled swimming organisms and the study of restricted chemotactic behavior. This research provides insights into how confinement affects bacterial chemotaxis and may inform the design of microfluidic devices for bacterial manipulation and analysis.

Bacteria frequently inhabit confined spaces such as soil pores, sediment interstices, rock crevices, biological tissues, and hydrogels. In these constrained environments, bacterial behavior often deviates significantly from that observed in unconfined conditions, typically resulting in reduced chemotactic ability. However, our findings challenge this notion, suggesting that certain types of

environmental confinement may actually enhance bacterial chemotaxis. This discovery could contribute to our understanding of microbial ecology across various habitats, including soil, oceans, and the human body.

Furthermore, these insights could have potential applications. In biotechnology, optimizing confinement conditions could improve the efficiency of bioremediation processes or the production of microbial products. In medicine, understanding how pathogens navigate confined spaces within the body could lead to novel strategies for preventing or treating infections.

Materials and methods

Strain and plasmids

The plasmid pTrc99a-mCherry expresses mCherry under an IPTG-inducible promoter. The wild-type *E. coli* K12 strain AW405 (HCB1) transformed with pTrc99a-mCherry was used for fluorescent tracking of cell trajectories. Cells were streaked on a Petri dish containing 1.5% agar and lysogeny broth (1% tryptone, 0.5% NaCl, 0.5% yeast extract, supplemented with the appropriate antibiotic: 100 µg/ml ampicillin). A single colony was inoculated into lysogeny broth and grown overnight at 33 °C with 200 rpm rotation.

The saturated culture was then diluted 1:100 (100 µl in 10 ml) into tryptone broth (1% tryptone, 0.5% NaCl, supplemented with the appropriate inducer and antibiotic: 0.1 mM IPTG, 100 µg/ml ampicillin) and grown at 33 °C with 200 rpm rotation to $OD_{600}=0.53$. Cells were then harvested from culture media by centrifugation at $1.2\times g$ for 6 min at room temperature. The pellet was resuspended by gently mixing in motility medium (10 mM potassium phosphate, 0.1 mM ethylenediaminetetraacetic acid (EDTA), 10 mM lactic acid, and 1 µM methionine at pH 7.0). The cells were washed three times to replace growth medium with motility medium. One hour after fluorescence maturation in a 33°C incubator, cells were placed in a 4°C refrigerator ready for microfluidic experiments.

Microfluidic device design and fabrication

A hydrogel-based microfluidic device containing lanes of different width was designed using L-edit based on previous work (42). The silicon mold with the positive relief features was fabricated using the standard soft lithography technique performed in our laboratory. Firstly, a 4-inch diameter silicon wafer was spin washed sequentially using water, acetone, methanol and isopropanol. The wafer was thoroughly dried on a 150 °C hot plate to ensure that organics has volatilized completely. Then, the wafer was treated with air plasma (Harrick plasma, PDC-002, 30W) for 10 min before the SU-8 2025 photoresist (Microchem) was spin-coated according to the products protocols. The spin-coated wafer was placed on a hot plate at 65°C for 3 min followed by 6 min at 95°C, then exposed to UV light using a maskless lithography machine (Heidelberg, µMLA) at 800 mJ/cm² with defocus was set to 0. The post-exposure bake was performed at 65°C for 2 min followed by 6 min at 95°C. Development procedure lasted about 6 minutes with constant shaking in the SU-8 developer. The wafer was rinsed with fresh developer followed by isopropanol. Finally, an approximately 30-µm-high photoresist layer was created as the master mold. The mold was fixed on a glass petri dish using double-sided tape.

Polydimethylsiloxane (PDMS) and curing agent was mixed with a 10:1 ratio (Sylgard 184, Dow Corning) and centrifuged at 3500 rpm for 5 min to remove air bubbles. The mixture was poured over the mold in glass petri dish and then placed at 4°C for an hour to avoid interference from very small air bubbles. The PDMS was cured at 80°C for 3 h, then separated from the wafer and cut into pieces, and holes were punched for inlets and outlets. The punched PDMS was cleaned with adhesive tape. The cleaned PDMS was bonded to a glass slide after air plasma treatment (Harrick plasma, PDC-002, 30W). The fabricated device was baked at 80°C overnight to establish covalent bonding.

The hydrogel agarose was used at 2% (w/v) concentration in motility medium to create diffusion-permeable walls. Agarose was slowly injected through the inlet port using a syringe pump at a constant flow rate of 2 $\mu\text{L}/\text{min}$, and the injection process was carried out in a drying oven at a constant temperature of 68 $^{\circ}\text{C}$.

Microscopy and data acquisition

The PDMS microfluidic chip was placed on a Nikon Ti-E inverted optical microscope equipped with a specific filter set, a 20 $\times$ objective (Nikon, CFI S Plan Fluor ELWD, 20 $\times$, NA 0.45), and a CMOS camera (Hamamatsu, ORCA-Flash4.0, pixel size = 6.5 μm). We set the objective lens's correction collar to 1.2 mm to match the glass slide thickness. A halogen lamp serves as the bright-field light source to help locate the observation area, while a xenon lamp serves as the excitation light source for fluorescence experiments. The depth of field is 3.7 μm .

1) Gradient profile calibration

Fluorescein solution (Sigma) was used for visualizing the chemical concentration field in the microfluidic device. 100 μM fluorescein and motility medium were flowed into the source and sink channels respectively, at a constant flow rate of 10 $\mu\text{L}/\text{min}$. One fluorescent image was captured per minute. To avoid the impact of photobleaching on the results, we only illuminated the sample during the imaging process.

2) Chemotaxis assays

The aspartic acid and motility medium solutions were respectively introduced into the source and sink channels, and left for more than 1 hour to allow aspartic acid to diffuse through the agarose wall and form a stable concentration gradient field. Then, bacteria were introduced into the cell channel at a rate of 5 $\mu\text{L}/\text{min}$ using a syringe pump for a duration of 90 seconds. After the bacteria injection was complete, the microvalves were closed to seal the cell channel. Bacteria then performed chemotaxis due to the perception of the concentration gradient field. After waiting for 2 minutes, the shutter was opened to illuminate the fluorescent bacteria with the excitation light, and video recording began. During the video acquisition process, fluorescence excitation light was continuously illuminated onto the bacteria. For each recording, 9600 frames (2048 $\times$ 2048 pixels) were captured at a frame rate of 20 fps and an exposure of 50 ms. Although the long-term illumination caused a decrease in the brightness of the fluorescent bacteria, the position information of the bacteria could still be accurately distinguished within the video duration (8 min).

Data analysis

Tracking

Image analysis was performed with the ImageJ (National Institutes of Health) TrackMate plugin. We chose the Log detector with an estimated blob diameter of 70 μm and a threshold of 800 μm to effectively recognize bacteria, based on the diameter of bacteria and the use of a 20 $\times$ objective. Then, we used the LAP Tracker and set the maximum distance for frame-to-frame linking to 50 μm .

Swimming speed

The velocity is calculated using the fourth order central difference method by (44):

$$\vec{v}_i = \frac{8(\vec{x}_{i+1} - \vec{x}_{i-1}) - (\vec{x}_{i+2} - \vec{x}_{i-2})}{12T}$$

The magnitude of the velocity is given by $v_i = |\vec{v}_i|$. The speed of each trajectory is defined as the average of v_i . The swimming speeds of bacteria in lanes of different widths are represented by the mean and standard deviation of all trajectories.

Calculation of the radius of circular motion on the solid surface

A total of 382 trajectories with a rotational exponent greater than 1.6 in the MA region of the $w = 44 \mu\text{m}$ lanes were selected for analysis. The trajectories were smoothed using the Savitzky-Golay filter with a window size of 5. The least squares method was then applied to solve the linear equations, yielding the center and radius of the fitted circles. The principle of this process is briefly described as follows:

The standard equation of a circle can be written as:

$$(x - a)^2 + (y - b)^2 = r^2,$$

where (a, b) are the coordinates of the circle's center, and r is the radius. Expanding and rearranging the standard equation of a circle, it can be rewritten as:

$$x^2 + y^2 = 2ax + 2by + c,$$

$$\text{where } c = r^2 - a^2 - b^2.$$

Rewriting the above equation in matrix form, we have:

$$B = A \cdot \text{Coefficients}$$

where

$$B = x^2 + y^2 = \begin{bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_n^2 + y_n^2 \end{bmatrix}$$

is the response vector, and

$$A = \begin{bmatrix} x & y & 1 \end{bmatrix} = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots \\ x_n & y_n & 1 \end{bmatrix}$$

is the design matrix. The coefficients are given by:

$$\text{Coefficients} = \begin{bmatrix} 2a \\ 2b \\ c \end{bmatrix} = (A^T A)^{-1} (A^T B).$$

Tumble angle

A tumble is defined as points where the velocity is lower than the average velocity of the trajectory $\langle v_i \rangle$ and shows a substantial drop from the previous time point. Specifically, a tumble is identified when:

$$\{v_{t1}, \dots, v_{tn}\} < \langle v_i \rangle,$$

$$\{v_{t0} - v_{t1}, \dots, v_{tn-1} - v_{tn}\} > \langle v_i \rangle \cdot \alpha,$$

where $n \geq 2$, $\alpha = 1/6$ is the threshold, and t_1 and t_n represent the start and end times of a tumble, respectively.

Simulation of bacterial chemotaxis within lanes

In our simulation, we introduced the effects of bottom surface and sidewalls into the 2-dimensional chemotaxis of *E. coli*. The simulations were carried out in a rectangular area of size $160 \mu\text{m} \times w$, where w denotes the lane width. Cells were treated as self-propelled particles. They could sense and adapt to the ligand in the surrounding environment via the intracellular chemotaxis signaling pathway, which could be described by a coarse-grained model of bacterial chemotaxis (45, 46):

$$a = \frac{1}{1 + \exp\left(N\left(\alpha(m - m_0) + \ln \frac{1 + \frac{c}{K_{off}}}{1 + \frac{c}{K_{on}}}\right)\right)},$$

$$\frac{dm}{dt} = k_R(1 - a) - k_B a,$$

where a , m , and c represent the activity of the receptor-kinase complex, the methylation level of receptors, and the ligand concentration, respectively. N denotes the number of chemoreceptors in a Monod-Wyman-Changeux cluster (47). The values of parameters we used were the same as before (4, 47, 48): $N = 4.6$, $K_{off} = 1.7 \mu\text{M}$, $K_{on} = 12 \mu\text{M}$, $\alpha = 1.7$, $m_0 = 1.0$, $k_R = 0.005 \text{ s}^{-1}$, $k_B = 0.010 \text{ s}^{-1}$. The concentration of CheY-P, which changes due to the receptor-kinase phosphorylation on CheY, could be calculated by $Y_p = 7.86a$. Consequently, the CW bias (B) of flagellar motors would also vary (49):

$$B = \frac{(Y_p)^{10.3}}{(Y_p)^{10.3} + 3.1^{10.3}}.$$

It has been reported that the average run time in the near-surface region for wildtype cells is $\sim 2.0 \text{ s}$ (corresponding to a run-to-tumble rate of $\sim 0.5 \text{ s}^{-1}$) (11), and the steady-state motor CW bias is ~ 0.15 without a gradient. The run-to-tumble rate k was set to be equivalent to the motor CCW-to-CW switching rate. As this rate increases linearly with motor bias B (50), it was calculated using $k = \frac{B}{0.31} \text{ s}^{-1}$. The switching rate from tumble to run was 5 s^{-1} due to the constant tumble duration of 0.2 s (51).

In this way, the motion states of each cell at any moment are determined. During a run, cells swim smoothly with a constant speed of $20 \mu\text{m/s}$, influenced by rotational diffusion and the effect of the bottom surface on bacterial swimming:

$$\theta(t + \Delta t) - \theta(t) = \left(-\frac{v_0}{r}\right)\Delta t + n(0, 2D_r\Delta t),$$

$$x(t + \Delta t) - x(t) = v_0 \cos \theta \Delta t,$$

$$y(t + \Delta t) - y(t) = v_0 \sin \theta \Delta t,$$

where θ is the angle between the swimming direction and the positive x -axis, $v_0 = 20 \mu\text{m/s}$ is the swimming speed, r is the radius of circular swimming due to the effect of the bottom surface, and $n(0, 2D_r\Delta t)$ represents a normal distribution with zero mean and variance $2D_r\Delta t$. The rotational diffusion coefficient was $0.062 \text{ rad}^2/\text{s}$ (52). During a tumble, cells stop swimming and a tumble angle is selected from the distribution that we measured (Fig. 2E). We also imposed a maximum detention time limit which, when combined with the variable tumble rate, results in an average wall residence time of approximately 2 s , matching our experimental observations (Fig. S6B). Cells will get a random y -coordinate at the other side when they swim through the x boundary.

Data availability

All data generated or analysed during this study are included in the manuscript and supporting files.

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Additional information

Author Contributions

J. Y., R.Z., and C. Z. designed the work. C. Z. and C. Y. performed the experiments. All authors wrote the manuscripts.

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MOST National Natural Science Foundation of China (NSFC)	12090053	Junhua Yuan
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Author ORCID iDs

Chi Zhang: <https://orcid.org/0000-0003-3154-2657>

Junhua Yuan: <https://orcid.org/0000-0002-6437-0655>

Additional files

[Supplemental materials](#) 

[Video S1](#) 

[Video S2](#) 

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Peer reviews

Reviewer #1 (Public review):

The authors show experimentally that, in 2D, bacteria swim up a chemotactic gradient much more effectively when they are in the presence of lateral walls. Systematic experiments identify an optimum for chemotaxis for a channel width of $\sim 8\mu\text{m}$, a value close to the average radius of the circle trajectories of the unconfined bacteria in 2D. These chiral circles impose that the bacteria swim preferentially along the right-side wall, which indeed yields chemotaxis in the presence of a chemotactic gradient. These observations are backed by numerical simulations and a geometrical analysis.

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Reviewer #3 (Public review):

This paper addresses, through experiment and simulation, the combined effects of bacterial circular swimming near no-slip surfaces and chemotaxis in simple linear gradients. The authors have constructed a microfluidic device in which a gradient of L-aspartate is established, to which bacteria respond while swimming while confined in channels of different widths. There is a clear effect that the chemotactic drift velocity reaches a maximum in channel widths of about 8 microns, similar in size to the circular orbits that would prevail in the absence of side walls. Numerical studies of simplified models confirm this connection.

The experimental aspects of this study are well executed. The design of the microfluidic system is clever in that it allows a kind of "multiplexing" in which all the different channel widths are available to a given sample of bacteria.

The authors have included a useful intuitive explanation of their results via a geometric model of the trajectories. In future work it would be interesting to analyze further the voluminous data on the trajectories of cells by formulating the mathematical problem in terms of a suitable Fokker-Planck equation for the probability distribution of swimming directions. In particular, this might help understand how incipient circular trajectories are interrupted by collisions with the walls and how this relates to enhanced chemotaxis.

The authors argue that these findings may have relevance to a number of physiological and ecological contexts. As these would be characterized by significant heterogeneity in pore sizes and geometries, further work will be necessary to translate the present results to those situations.

<https://doi.org/10.7554/eLife.102686.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

This article deals with the chemotactic behavior of E coli bacteria in thin channels (a situation close to 2D). It combines experiments and simulations.

The authors show experimentally that, in 2D, bacteria swim up a chemotactic gradient much more effectively when they are in the presence of lateral walls. Systematic experiments identify an optimum for chemotaxis for a channel width of $\sim 8\mu\text{m}$, close to the average radius of the circle trajectories of the unconfined bacteria in 2D. It is known that these circles are chiral and impose that the bacteria swim preferentially along the right-side wall when there is no chemotactic gradient. In the presence of a chemotactic gradient, this larger proportion of bacteria swimming on the right wall yields chemotaxis. This effect is backed by numerical simulations and a geometrical analysis.

If the conclusions drawn from the experiments presented in this article seem clear and interesting, I find that the key elements of the mechanism of this wall-directed chemotaxis are not sufficiently emphasized. Moreover, the paper would be clearer with more details on the hypotheses and the essential ingredients of the analyses.

We thank the reviewer for these constructive suggestions. We agree that emphasizing the underlying mechanism is crucial for the clarity of our findings. In the revised manuscript, we

have now explicitly highlighted the critical roles of chiral circular motion and the alignment effect following side-wall collisions in both the Abstract (lines 25-27) and the Discussion (lines 391-393). Furthermore, we have added a new analysis of bacterial trajectories post-collision (Fig. S2), which demonstrates that cells predominantly align with and swim along the sidewalls. We have also clarified the assumptions in our numerical simulations, specifically how the radius of circular trajectories and the alignment effect are incorporated into the equations of motion. Please refer to our detailed responses in the "Recommendations for the authors" section for further specifics.

Reviewer #2 (Public review):

Summary:

In this study, the authors investigated the chemotaxis of E. coli swimming close to the bottom surface in gradients of attractant in channels of increasingly smaller width but fixed height = 30 μm and length $\sim 160 \mu\text{m}$. In relatively large channels, they find that on average the cells drift in response to the gradient, despite cells close to the surface away from the walls being known to not be chemotactic because they swim in circles.

They find that this average drift is due to the cell localization close to the side walls, where they slide along the wall. Whereas the bacteria away from the walls have no chemotaxis (as shown before), the ones on the left side wall go down-gradient on average, but the ones on the right-side wall go up-gradient faster, hence the average drift. They then study the effect of reducing channel width. They find that chemotaxis is higher in channels with a width of about 8 μm , which approximately corresponds to the radius of the circular swimming R. This higher chemotactic drift is concomitant to an increased density of cells on the RSW. They do simulations and modeling to suggest that the disruption of circular swimming upon collision with the wall increases the density of cells on the RSW, with a maximal effect at $w = \sim 2/3 R$, which is a good match for their experiments.

Strengths:

The overall result that confinement at the edge stabilises bacterial motion and allows chemotaxis is very interesting although not entirely unexpected. It is also important for understanding bacterial motility and chemotaxis under ecologically relevant conditions, where bacteria frequently swim under confinement (although its relevance for controlling infections could be questioned). The experimental part of the study is nicely supported by the model.

Weaknesses:

Several points of this study, in particular the interpretation of the width effect, need better clarification:

(1) Context:

There are a number of highly relevant previous publications that should have been acknowledged and discussed in relation to the current work:

<https://pubs.rsc.org/en/content/articlehtml/2023/sm/d3sm00286a> 

<https://link.springer.com/article/10.1140/epje/s10189-024-00450-7> 

<https://doi.org/10.1016/j.bpj.2022.04.008> 

<https://doi.org/10.1073/pnas.1816315116> 

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We appreciate the reviewer bringing these important publications to our attention. We have now cited and discussed these works in the Introduction (lines 55-62 and 76-85) to better contextualize our study regarding bacterial motility and chemotaxis in confined geometries.

(2) *Experimental setup:*

a) The channels are built with asymmetric entrances (Figure 1), which could trigger a ratchet effect (because bacteria swim in circle) that could bias the rate at which cells enter into the channel, and which side they follow preferentially, especially for the narrow channel. Since the channel is short (160 μm), that would reflect on the statistics of cell distribution. Controls with straight entrances or with a reversed symmetry of the channel need to be performed to ensure that the reported results are not affected by this asymmetry.

We appreciate the reviewer's insight regarding the potential ratchet effect caused by asymmetric entrances. To rule this out, we fabricated a control device with straight entrances and repeated the measurements. As shown in Figure S3, the chemotactic drift velocity follows the same trend as observed in the original setup, confirming an optimal width of ~ 9 mm. These results demonstrate that the entrance geometry does not bias the reported statistics. We have updated the manuscript text at lines 233-235.

b) The authors say the motile bacteria accumulate mostly at the bottom surface. This is strange, for a small height of 30 μm , the bacteria should be more-or-less evenly spread between the top and bottom surface. How can this be explained?

We apologize for not explaining this clearly in the text. As shown by Wei et al., Phys. Rev. Lett. 135, 188401 (2025), significant surface accumulation occurs in channels with heights exceeding 20 μm . In our specific experimental setup, we did not use Percoll to counteract gravity. Therefore, the bacteria accumulated mostly at the bottom surface under the combined influence of gravity and hydrodynamic attraction. This bottom-surface localization is supported by our observation that the bacterial trajectories were predominantly clockwise (characteristic of the bottom surface) rather than counter-clockwise (characteristic of the top surface). We have added this explanation to Line 141.

c) At the edge, some of the bacteria could escape up in the third dimension (<http://doi.org/10.1039/c5sm00939a>). What is the magnitude of this phenomenon in the current setup? Does it have an effect?

We thank the reviewer for raising this important point regarding 3D escape. We have quantified this phenomenon and found the escape rate from the edge into the third dimension to be 0.127 s^{-1} . This corresponds to a mean residence time that allows a cell moving at 20 mm/s to travel approximately 157.5 mm along the edge. Since this distance is comparable to the full length of our lanes (~ 160 mm), most cells traverse the entire edge without escaping. Furthermore, our analysis is based on the average drift of the surface trajectories per unit of time; this metric is independent of the absolute number of cells present. Therefore, the escape phenomenon does not significantly impact our conclusions. We have added a statement clarifying this at line 154.

d) What is the cell density in the device? Should we expect cell-cell interactions to play a role here? If not, I would suggest to de-emphasize the connection to chemotaxis in the

swarming paper in the introduction and discussion, which doesn't feel very relevant here, and rather focus on the other papers mentioned in point 1.

The cell density in our experiments was approximately $1.3 \times 10^{-3} \mu\text{m}^{-2}$. Given this low density, we do not expect cell-cell interactions to play a role in the observed behaviors.

Regarding the connection to swarming chemotaxis: We agree that our low-density setup differs from a high-density swarm; however, we believe the comparison remains relevant for two reasons. First, it provides a necessary contrast to studies showing surface inhibition of chemotaxis. Second, while we eliminate cell-cell interactions, we isolate the geometric aspect of swarming. In a swarm, cells move within narrow lanes created by their neighbors. Our device mimics this specific physical confinement by replacing neighboring cells with PDMS sidewalls. This allows us to decouple the effects of physical confinement from cell-cell interactions. We have added the text (Line 370) to clarify this rationale and have incorporated the additional references in introduction as suggested in point 1.

e) We are not entirely convinced by the interpretation of the results in narrow channels. What is the causal relationship between the increased density on the RSW and the higher chemotactic drift? The authors seem to attribute higher drift to this increased RSW density, which emerges due to the geometric reasons. But if there is no initial bias, the same geometric argument would induce the same increased density of down-gradient swimmers on the LSW, and so, no imbalance between RSW and LSW density. Could it be the opposite that the increased RSW density results from chemotaxis (and maybe reinforces it), not the other way around? Confinement could then deplete one wall due to the proximity of the other, and/or modify the swimming pattern - $8 \mu\text{m}$ is very close to the size of the body + flagellum. To clarify this point, we suggest measuring the bacterial distributions in the absence of a gradient for all channel widths as a control.

We thank the reviewer for this insightful comment regarding the causal relationship between cell density and chemotactic drift. We apologize if the initial explanation was unclear.

Regarding the no-gradient control: Without an attractant gradient (and no initial bias), there is no breaking of symmetry and the labels of "LSW" and "RSW" are arbitrary. Therefore, there will be no asymmetry in the bacterial distributions on both sides (within experimental fluctuations) in the absence of a gradient for any channel width.

Regarding the causality and density imbalance: We agree that the increased RSW density is a result of chemotaxis, which is then reinforced by the lane geometry especially at narrow lane width. The mechanism relies on the coupling of chemotactic bias with surface circularity. The angle ranges that lead to RSW-UG accumulation (Fig. 6A-C) coincide with the up-gradient direction. Because these cells experience suppressed tumbling (longer runs), they can maintain the steady circular trajectories required to reach and align with the RSW. Conversely, while pure geometric analysis suggests a similar potential for LSW-DG accumulation, these trajectories coincide with the down-gradient direction. These cells experience enhanced tumbling, which distorts the circular trajectories. This prevents them from effectively reaching the LSW and also increases the probability of them leaving the wall. Therefore, the causality is indeed a positive feedback loop: the attractant gradient creates an initial bias that allows the RSW-UG fraction to form stable trajectories; the optimal lane width (matching the swimming radius) then maximizes this capture efficiency, further enriching the RSW fraction and enhancing the overall drift.

We have added clarifications regarding these points in the revised manuscript (the last paragraph of "Results").

(3) Simulations:

The simulations treat the wall interaction very crudely. We would suggest treating it as a mechanical object that exerts elastic or "hard sphere" forces and torques on the bacteria for more realistic modeling.

We appreciate the reviewer's suggestion to incorporate more detailed mechanical interactions, such as elastic or hard-sphere forces, for the wall collisions. While we agree that a full hydrodynamic or mechanical model would offer higher fidelity, our experimental observations suggest that a simplified kinematic approach is sufficient for the specific phenomena studied here.

As shown in the new Fig. S2, our analysis of cell trajectories in the 44- μm -wide channels reveals that cells colliding with the sidewalls tend to align with the surface almost instantaneously. The timescale required for this alignment is negligible compared to the typical wall residence time (see also Ref. 6). Consequently, to maintain computational efficiency without sacrificing the essential physics of the accumulation effect, we employed a coarse-grained phenomenological model where a bacterium immediately aligns parallel to the wall upon contact, similar to approaches used previously (Ref. 43). We have added relevant text to the manuscript on lines 168-171.

Notably, the simulations have a constant (chemotaxis independent) rate of wall escape by tumbling. We would expect that reduced tumbling due to up-gradient motility induces a longer dwell time at the wall.

We apologize for the confusion. The chemotaxis effect is indeed fully integrated into our simulation. Specifically, the simulated cells sense the chemical gradient and adjust their motor CW bias (B) accordingly. This adjustment directly modulates the tumble rate (k), calculated as $k = B/0.31 \text{ s}^{-1}$. Consequently, the wall escape rate is not constant but varies with the chemotactic response. We also imposed a maximum detention time limit which, when combined with the variable tumble rate, results in an average wall residence time of approximately 2 s, consistent with our experimental observations (Fig. S6B). We have clarified these details in the final section of 'Materials and Methods'.

Reviewer #3 (Public review):

This paper addresses through experiment and simulation the combined effects of bacterial circular swimming near no-slip surfaces and chemotaxis in simple linear gradients. The authors have constructed a microfluidic device in which a gradient of L-aspartate is established to which bacteria respond while swimming while confined in channels of different widths. There is a clear effect that the chemotactic drift velocity reaches a maximum in channel widths of about 8 microns, similar in size to the circular orbits that would prevail in the absence of side walls. Numerical studies of simplified models confirm this connection.

The experimental aspects of this study are well executed. The design of the microfluidic system is clever in that it allows a kind of "multiplexing" in which all the different channel widths are available to a given sample of bacteria.

While the data analysis is reasonably convincing, I think that the authors could make much better use of what must be voluminous data on the trajectories of cells by formulating the mathematical problem in terms of a suitable Fokker-Planck equation for the probability distribution of swimming directions. In particular, I would like to see much more analysis of how incipient circular trajectories are interrupted by collisions with the walls and how this relates to enhanced chemotaxis. In essence, there needs to be a much clearer control analysis of trajectories without sidewalls to understand the mechanism in their presence.

We thank the reviewer for this insightful suggestion. We agree that understanding how circular trajectories are interrupted by wall collisions is central to explaining the enhanced chemotaxis. While we did not explicitly formulate a Fokker-Planck equation, we have addressed the reviewer's core point by employing two complementary mathematical approaches that model the probability distribution of swimming directions and wall interactions:

(1) Stochastic simulations (Langevin approach): As detailed in the "Simulation of *E. coli* chemotaxis within lane confinements" subsection of "Results" and Figure 5, we modeled cells as self-propelled particles performing random walks. This model explicitly accounts for the "interruption" of circular trajectories by incorporating a constant angular velocity (circular swimming) and an alignment effect upon collision with sidewalls. These simulations successfully reproduced the experimental trends, confirming that the interplay between circular radius and lane width determines the optimal drift velocity.

(2) Geometric probability analysis: To provide the "intuitive understanding", we included a specific Geometrical Analysis section (the last subsection of "Results") and Figure 6. This analysis mathematically formulates the problem by calculating the exact proportion of swimming angles that allow a cell to transition from a circular trajectory in the bulk to an up-gradient trajectory along the Right Sidewall (RSW). By integrating over the possible swimming directions, we derived the probability of wall interception as a function of lane width (w) and swimming radius (r). This analysis reveals that the interruption of circular paths is most favorable for chemotaxis when $w \gg (0.7-0.8)r$.

(3) Control analysis: regarding the "control analysis of trajectories without sidewalls," we utilized the cells in the Middle Area (MA) of the wide lanes as an internal control. As shown in Fig. 2B and 4A, these cells exhibit typical surface-associated circular swimming (Fig. 3B) but generate zero net drift. This serves as the baseline "no sidewall" condition, demonstrating that the chemotactic enhancement is strictly driven by the rectification of circular swimming into wall-aligned motion at the boundaries.

The authors argue that these findings may have relevance to a number of physiological and ecological contexts. Yet, each of these would be characterized by significant heterogeneity in pore sizes and geometries, and thus it is very unclear whether or how the findings in this work would carry over to those situations.

We thank the reviewer for this important observation regarding environmental heterogeneity. We agree that we should be cautious about directly extrapolating to complex ecological contexts without qualification. We have revised the last sentence of the abstract to adopt a more measured tone: "Our results may offer insights into bacterial navigation in complex biological environments such as host tissues and biofilms, providing a preliminary step toward exploring microbial ecology in confined habitats and potential strategies for controlling bacterial infections."

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Key elements of the mechanism of wall-directed chemotaxis are not sufficiently emphasized:

For instance, the chirality of the trajectories is an essential part of the analysis but is mentioned only briefly in the introduction. In the geometrical analysis, I understand that one of the critical parameters is the angle at which bacteria "collide" with the walls. But, again, this remains largely implicit in the discussion. This comes to the point that these ideas are not even mentioned in the abstract which doesn't provide any hint of a

mechanism. An analysis of the actual trajectories of the cells after they hit the walls, as a function of their initial angle would be helpful in comparison with the simulations and the geometrical analysis.

We appreciate the reviewer's insightful comment regarding the need to better emphasize the mechanism of wall-directed chemotaxis. We agree that the chirality of trajectories and the geometry of wall collisions are central to our analysis and were previously under-emphasized.

To address this, we have made the following revisions:

(1) We have revised the Abstract (lines 25-27) and the Discussion (lines 391-393) to explicitly highlight the crucial role of chiral circular motion and the alignment effect following sidewall collisions.

(2) We further analyzed bacterial trajectories at different collision angles. Typical examples are shown in Supplementary Fig. S2. We observed that cells tend to align with and swim along the sidewalls regardless of their initial collision angles. This finding is now described in the main text at lines 168-171.

The motion of the bacteria is modelled as run-and-tumble at several places in the manuscript, and in particular in the simulations. Yet, the trajectories of the bacteria seem to be smooth in this almost 2D geometry, except of course when they directly interact with the walls (I hardly see tumbles in the MA region in Figure 1B). Can the authors elaborate on the assumptions made in the numerical simulations? In particular, how is the radius of the trajectories included in these equations of motion (line 514)?

We apologize for the lack of clarity regarding the bacterial motion model. It has been established that while bacteria do tumble near solid surfaces, they exhibit a smaller reorientation angle compared to bulk fluids; in fact, the most probable reorientation angle on a surface is zero (Ref. 41). Consequently, tumbles are often difficult to distinguish from runs with the naked eye. Additionally, the trajectories in Figure 1B are plotted on a 44 mm × 150 mm canvas with unequal coordinate scales, which may further obscure the visual distinctness of tumbling events.

Regarding the equations of motion: We modeled the bacteria as self-propelled particles governed by the internal chemotaxis pathway, alternating between run and tumble states. As noted in the equations on lines 286 & 578, we incorporated the circular motion by introducing a constant angular velocity, $-\nu_0/r$, during the run state. Here, ν_0 represents the swimming speed, r denotes the radius of circular swimming, and the negative sign indicates clockwise chirality. Furthermore, to model the hydrodynamic interaction with the boundaries, we assumed that when a cell collides with a sidewall, its velocity vector instantly aligns parallel to that wall.

The comparison of Figure 5B (simulations) with Figure 4B (experiments) does not strike me as so "similar". Why are the points at small widths so noisy (Figure 5AB)? Figure 5C is cut at these widths, it should be plotted over the entire scale.

We acknowledge that the agreement between simulation and experiment is less robust in the narrowest channels. The discrepancy and "noise" at small widths in Figure 5 arise from the limitations of the self-propelled particle model in highly confined geometries. Specifically, our simulation treats bacteria as point particles and does not explicitly calculate the physical exclusion (steric effects) caused by the finite size of the flagella and cell body.

In the experimental setup, steric constraints within narrow channels (comparable to the cell size) restrict the cells' ability to turn freely, effectively stabilizing their motion. However, because our model allows particles to reorient more freely than actual cells would in such

confined spaces, it produces fluctuations and an overestimation of the drift velocity at small widths. If these confinement effects were fully incorporated, the cell density mismatch between the left and right sidewalls would be reduced, leading to lower drift velocities that match the experimental data more closely.

Regarding Figure 5C: Since the "active particle" assumption loses physical validity in channels narrower than the scale of the bacterium, the simulation results in this regime are not representative of biological reality. Plotting these non-physical points would distort the analysis. Therefore, we have maintained the truncation of Figure 5C at 4 μm to ensure the data presented is physically meaningful. We have added a clear discussion of these model limitations to the manuscript at lines 310-314.

These important precisions should be added to the text or in a supplementary section. A validated mechanism describing in detail the impact of the walls on the cell trajectories would greatly improve the conclusions.

We thank the reviewer for the suggestions. As noted in the responses above, we have incorporated the details concerning the simulation assumptions and the model limitations at narrow widths into the revised manuscript. We have performed further analysis of the collision trajectories between bacteria and the sidewalls. As illustrated in the new Fig. S2, the data confirms that cells tend to align with and swim along the sidewalls following a collision, regardless of the initial impact angle.

Reviewer #2 (Recommendations for the authors):

Minor points

(1) Related to swimming in 3D: The authors should specify the depth of field of the objective in their setup.

We thank the reviewer for pointing this out. We have calculated the depth of field (DOF) of our objective to be approximately 3.7 μm . This estimate is based on the standard formula:

$$DOF = \frac{\lambda \cdot n}{NA^2} + \frac{n}{M \cdot NA} \cdot e$$

where $\lambda = 610$ nm (emission wavelength), $n = 1.0$ (refractive index), $NA = 0.45$ (numeric aperture), $M = 20$ (magnification), and $e = 6.5$ μm (camera resolution). We have added this specification to the "Microscopy and Data Acquisition" section of "Materials and Methods".

(2) Related to the interpretation of the width effect: We think plotting the cell enrichment, ie the probabilities P in Figure 4B normalized to the expected value if cells were homogeneously distributed ($(3\mu\text{m})/w$ for the side walls, $(w - 6\mu\text{m})/w$ for the middle) would help understand the strength of the wall 'siphoning' effect.

We thank the reviewer for the suggestion. We have calculated the cell enrichment by normalizing the observed probabilities against the expected values for a homogeneous distribution, as suggested. The resulting relationship between cell enrichment and lane width is presented in Figure S4.

Related to simulations:

(1) Showing v_d for the 3 regions in Figure S5 would be helpful also to understand the underlying mechanism.

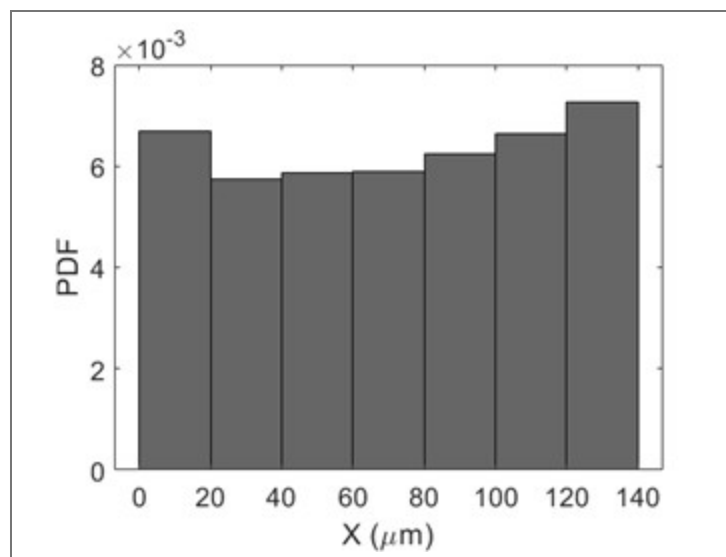
We thank the reviewer for the suggestion. The V_d values for the three regions are shown in Fig. S5.

(2) Figure 5B vs 4B: There is a mismatch in the right vs left side density at $w=6\mu\text{m}$ in the simulations that is not here in the experiments. What could explain this difference?

We appreciate the reviewer pointing this out. The mismatch in the simulations is due to the simplified treatment of cells as self-propelled particles, which overlooks the physical volume of the cell body and flagella. In narrow channels ($w=6\text{ mm}$), these physical constraints would restrict the cells' ability to change direction freely - a factor not fully captured in the simulation. Accounting for these steric effects would trap cells more effectively against the walls, reducing the density asymmetry between the LSW and RSW and lowering the drift velocity. This would bring the simulation results closer to the experimental observations. We have added a discussion of these limitations and effects to the revised manuscript (lines 310-314).

(3) The simulations essentially assume that the density of motile cells is homogeneous and equal at both $x=0$ and $x=L$ open ends of the channel. Is it the case in the experiments, even with the gradient, and the walls creating some cell transport?

We thank the reviewer for pointing this out. The simulation assumption is consistent with our experimental observations. Our data were recorded within $160\text{-}\mu\text{m}$ -long lanes located in the center of the wider ($400\text{ }\mu\text{m}$) cell channel. In this central region, the cells maintain a continuous flux. Furthermore, experiments were performed within 8 min of flow, limiting the time for significant cell density gradients to establish. As illustrated in Author response image 11, the inhomogeneity in the measured cell density distribution is insignificant across the length of the observation window, indicating that the walls and gradient do not create significant heterogeneity at the boundaries of the region of interest.



Author response image 1. The cell density distribution along the gradient field from the data of $44\text{-}\mu\text{m}$ -wide lane.

(4) Line 506: There is something strange with the definition of the bias. B cannot be the tumbling bias if $k=B/0.31\text{ s}^{-1}$ and the tumble-to-run rate is $5/\text{s}$, because then the tumbling bias is $B/0.31 / (B/0.31 + 5)$. Please clarify.

We apologize for the confusion caused by the notation. In our model, B represents the CW bias of the individual flagellar motor, not the macroscopic tumbling bias of the cell. We assume the run-to-tumble rate is equivalent to the motor CCW-to-CW switching rate (k).

Previous studies have shown that this rate increases linearly with the motor CW bias according to $k=B/t$, where t is a characteristic time (Ref. 50).

Based on experimental data for wildtype cells, the average run time in the near-surface region is ~ 2.0 s (corresponding to a run-to-tumble rate of ~ 0.5 s⁻¹) (Ref. 11), and the steady-state wildtype CW bias is ~ 0.15 . Using these values, we determined $t \sim 0.31$ s. Consequently, the switching rate is defined as $k=B/0.31$ s⁻¹. Since the tumble duration is constant (0.2 s) (Ref. 51), the tumble-to-run rate is fixed at 5 s⁻¹. We have clarified these definitions and parameter values in lines 569-573.

Other minor comments:

(1) Line 20 and lines 34-35: We think that the connection to infection is questionable here and should be toned down.

Thank you for the suggestion. We have revised Line 20 to read: "Understanding bacterial behavior in confined environments is helpful to elucidating microbial ecology and developing strategies to manage bacterial infections." Additionally, we modified lines 34-35 to state: "Our results may offer insights into bacterial navigation in complex biological environments such as host tissues and biofilms, providing a preliminary step toward exploring microbial ecology in confined habitats and potential strategies for controlling bacterial infections."

(2) Line 49: Consider highlighting the change in the sense of rotation at the air-liquid interface.

Thank you for the suggestion. We have now highlighted the difference in chirality between trajectories at the air-liquid interface and those at the liquid-solid interface. The text has been updated to read: "For example, *E. coli* swim clockwise when observed from above a solid surface, whereas *Caulobacter crescentus* move in tight, counter-clockwise circles when viewed from the liquid side."

(3) Lines 58-59: The sentence should be better formulated, explaining what is CheY-P and that its concentration changes because of a change in phosphorylation (P).

Thank you for the suggestion. We have reformulated this section to explicitly define CheY-P and explain how its concentration is regulated through phosphorylation. The revised text reads: "The transmembrane chemoreceptors detect attractants or repellents and transmit signals into the cell by modulating the autophosphorylation of the histidine kinase CheA. Attractant binding suppresses CheA autophosphorylation, while repellent binding promotes it. This modulation alters the concentration of the phosphorylated response regulator protein, CheY-P."

(4) Lines 63-64: CheR CheB do a bit more than "facilitating" adaptation, they mediate it. The notation CheB(p) may be confusing, since "-P" was used above for CheY.

Thank you for pointing this out. We have corrected the notation and strengthened the description of the enzymes' roles. The revised text is: "The adaptation enzymes CheR and CheB methylate and demethylate the receptors, respectively, mediating sensory adaptation."

(5) Line 130: there must be a typo in the formula.

We have replaced the ambiguous lag time variable in Fig. 1C with $n\Delta t$ to ensure mathematical consistency.

(6) Additionally, Δt is both the time between the frame here and the lag time in Figure 1.

Thank you for highlighting this ambiguity. We have updated the notation to distinguish these two values. The lag time in Figure 1 is now explicitly denoted as $n\Delta t$, while Δt remains the time interval between individual frames.

| (7) Line 162: "Consistent with previous reports," a reference to said reports is missing.

Thank you for pointing this out. We have now added the reference (Ref. 41) to support this statement.

| (8) Figure 1B: Are these tracks in the presence of a gradient? Same as used in panel C? This needs to be explained.

Response: Thank you for this question. We confirm that the tracks shown in Figure 1B were indeed recorded in the presence of a gradient and represent a subset of the data used in Figure 1C. We have clarified this in the figure legend as follows: "Thirty bacterial trajectories selected from the data of the 44-mm-wide lane in gradient assays. These represent a subset of the trajectories analyzed in panel C."

| (9) Simulations: the equation for $x(t)$ should also be given for completeness.

Thank you for the suggestion. For completeness, we have added the position updating equations for the run state to the Materials and Methods section (lines 579-580). The equations are defined as:

$$\begin{aligned}x(t + \Delta t) - x(t) &= v_0 \cos \theta \Delta t, \\y(t + \Delta t) - y(t) &= v_0 \sin \theta \Delta t.\end{aligned}$$

| (10) Figure S2: For the swimming directions that are more unstable due to the surface friction torque, RSW-DG, and LSW-UG, one would have expected that the Up-gradient motion is more persistent than the down gradient one. It seems to be the opposite. Is it significant, and what could be the reason for this?

We apologize for the lack of clarity in our original explanation. While we would generally expect up-gradient motion to be more persistent than down-gradient motion in bulk fluid, our measurements near the surface show a different trend due to the specific contributions of run and tumble states to the escape rate. Cells swimming up-gradient (UG) in the LSW experience higher probability of running. Consequently, they are subjected to the destabilizing surface friction torque for a greater proportion of time compared to cells swimming down-gradient (DG) in the RSW. This can be explained mathematically. The escape rates for RSW-DG and LSW-UG can be expressed as:

$$\begin{aligned}k_{RSW-DG} &= B^- k_T + (1 - B^-) k_R \\k_{LSW-UG} &= B^+ k_T + (1 - B^+) k_R\end{aligned}$$

Where B^+ and B^- represent the tumble bias (probability of tumbling) when swimming up-gradient and down-gradient, respectively, and k_T and k_R denote the escape rates during a tumble and a run, respectively. Due to the chemotactic response, $0 \leq B^+ < B^- \leq 1$. Crucially, our system is characterized by $k_R > k_T$ (the escape rate is higher during a run than a tumble). Therefore, the lower tumble bias during up-gradient swimming ($B^+ < B^-$) increases the weight of the run-state escape term ($(1 - B^+) k_R$), leading to a higher overall escape rate for LSW-UG compared to RSW-DG. We have added an intuitive understanding of $k_R > k_T$ in the Supplemental text.

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